

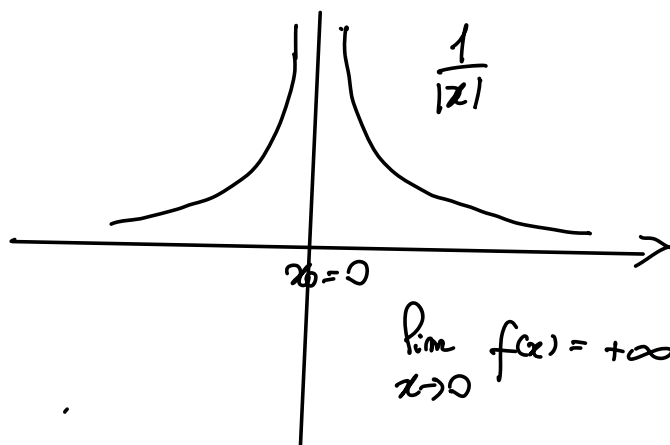
CASI (2)-(3)

$c = x_0 \in \mathbb{R}$ (di acc. per dom f)

$l = +\infty / -\infty$

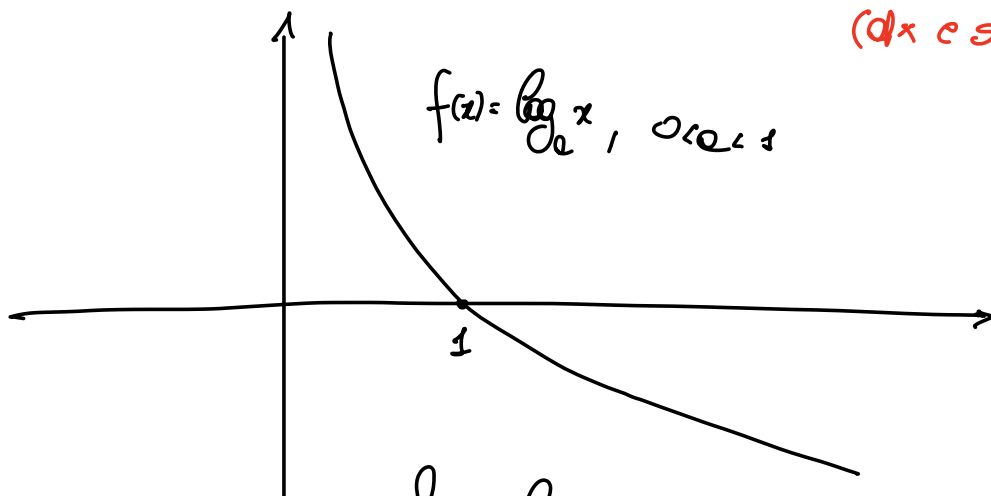
$\downarrow$
 $\lim_{x \rightarrow x_0} f(x) = +\infty$ (d'meno)
 sup. illimitato
 $\downarrow$
 $\lim_{x \rightarrow x_0} f(x) = -\infty$ (d'meno)
 inf. illimitato

$$\lim_{x \rightarrow x_0} f(x) = +\infty$$



$$\lim_{x \rightarrow 0} f(x) = +\infty$$

$x=0$ asintoto verticale
(d'x e s'x) completo



$$\lim_{x \rightarrow 0} \log_a x = +\infty$$

$$= \lim_{x \rightarrow 0^+} \log_a x$$

$\forall I(l) \exists I(c) \text{ t.c.}$

$\forall x \in \text{dom} f \cap I(c) \setminus \{c\} \Rightarrow f(x) \in I(c)$

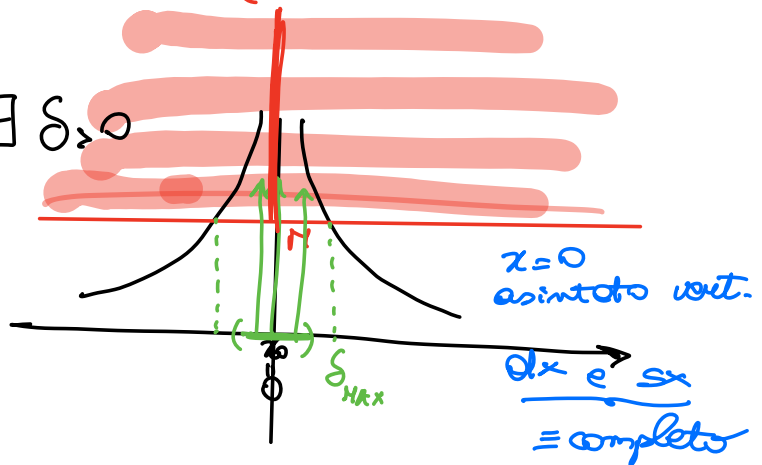
$$c = x_0 \in \mathbb{R} \Rightarrow I(c) = I(x_0) = (x_0 - \delta, x_0 + \delta)$$

$$I(\ell) = I(+\infty) = (M, +\infty) \quad , \quad M \in \mathbb{R}$$

$$\forall I(\ell) = I(+\infty) = (M_1 + \infty) \quad \text{and} \quad \forall M \in \mathbb{R}$$

(antici da un test: $H/H > 0$)

$$\exists \bar{I}_\delta(x_0) \text{ and } \exists \delta_0$$



f.c.

$$\underbrace{\forall x \in \text{dom} f \cap I_\delta(x_0) \setminus \{x_0\}}_{\text{ii)}} \Rightarrow \underbrace{f(x) \in I(+\infty)}_{f(x) > M}$$

$$\forall z \in \text{dom} f, 0 < |z - z_0| < \delta$$

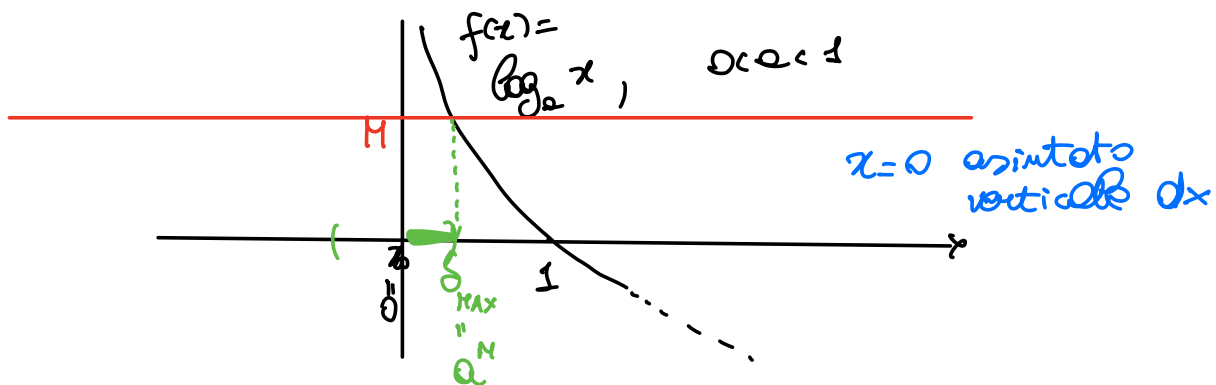
quindi

CASO 2

$$\lim_{x \rightarrow \infty} f(x) = +\infty$$

$$\forall M \in \mathbb{R} \quad (\text{and: } \forall M > 0) \quad \exists \delta > 0 \text{ t.c.}$$
$$\forall x \in \text{dom} f, 0 < |x - x_0| < \delta \Rightarrow f(x) > M$$

(ES) $\lim_{x \rightarrow 0} \log x = +\infty$



Fino $M \in \mathbb{R}$ a piacere, denotare $\delta > 0$ t.c.

$$\forall x \in \underbrace{\text{dom} f}_{(0, +\infty)}, x \neq 0, |x-0| < \delta \Rightarrow f(x) > M$$

parts $\log_a x > M \Rightarrow a^{y_1} < a^{y_2} \Rightarrow a^{y_1} < a^{y_2}$

N.B. $0 < a < 1 \Rightarrow \lim_{n \rightarrow \infty} a^n = 0 \Rightarrow a^x \downarrow$
 inverse
 $a^x, 0 < a < 1$

$$0 < \delta < 1, \quad \log_a x > M \Rightarrow x < a^M$$

$\equiv \delta_{MAX}$

Abbiamo dimostrato:

$\forall M \in \mathbb{R}, \exists \delta > 0$ ($\delta_{MAX} = a^M$, ma vanno bene anche tutti i $0 < \delta \leq \delta_{MAX}$) t.c.

$$\forall x \in (0, \delta_{MAX} = a^M) \Rightarrow f(x) = \log_a x > M$$

N.B. $\lim_{x \rightarrow 0} \log_a x = \lim_{x \rightarrow 0^+} \log_a x$



CASO (3)

$$\lim_{x \rightarrow x_0} f(x) = -\infty$$

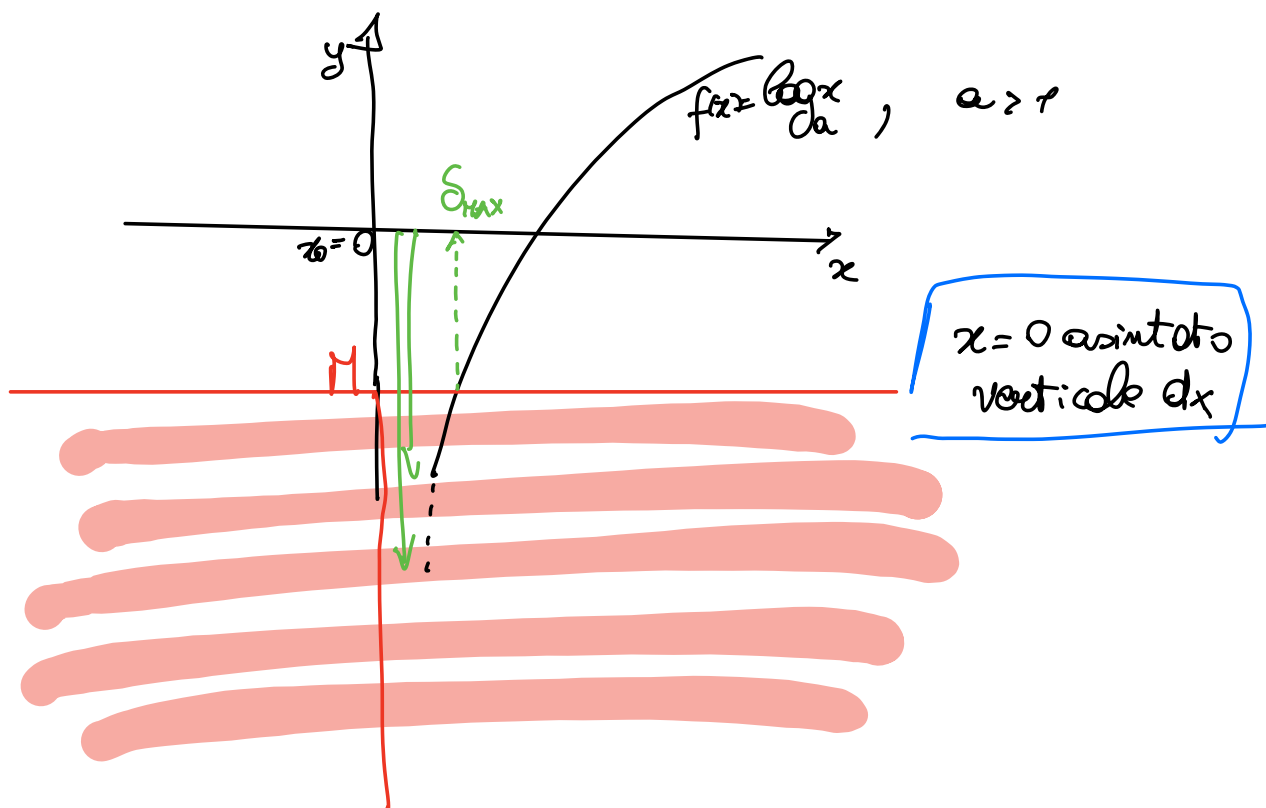
$$I(-\infty) = (-\infty, M)$$

(anzi $\forall M < 0$)

$$\forall M \in \mathbb{R} \quad \exists \delta > 0 \quad \text{t.c.}$$

$$\forall x \in \text{dom} f, 0 < |x - x_0| < \delta \Rightarrow f(x) < M$$

(ES)



$\forall M \in \mathbb{R} \quad \exists \delta > 0 \text{ t.c.}$

$$\forall x \in \text{dom} f, x \neq 0, |x - 0| < \delta \Rightarrow \log_a x < M$$

$$\underbrace{a > 1 \quad \log_a \uparrow, a^x \uparrow}_{x < a^M = \delta_{M, a}}$$

adesso abbiamo dimostrato

$$\forall M \in \mathbb{R} \quad \exists \delta^0 (\text{ad es. } \delta_{M, a} = a^M)$$

$$\text{t.c.} \quad \forall x \in (0, a^M) \Rightarrow f(x) = \log_a x < M$$

CASI ④ + ⑦ $\text{dom} f$ sup illimitato / inf illimitato

$$c = +\infty (-\infty), \quad l \in \mathbb{R}$$

$$\lim_{x \rightarrow +\infty} f(x) = l \quad (\Rightarrow y = l \text{ asintoto orizzontale dx})$$

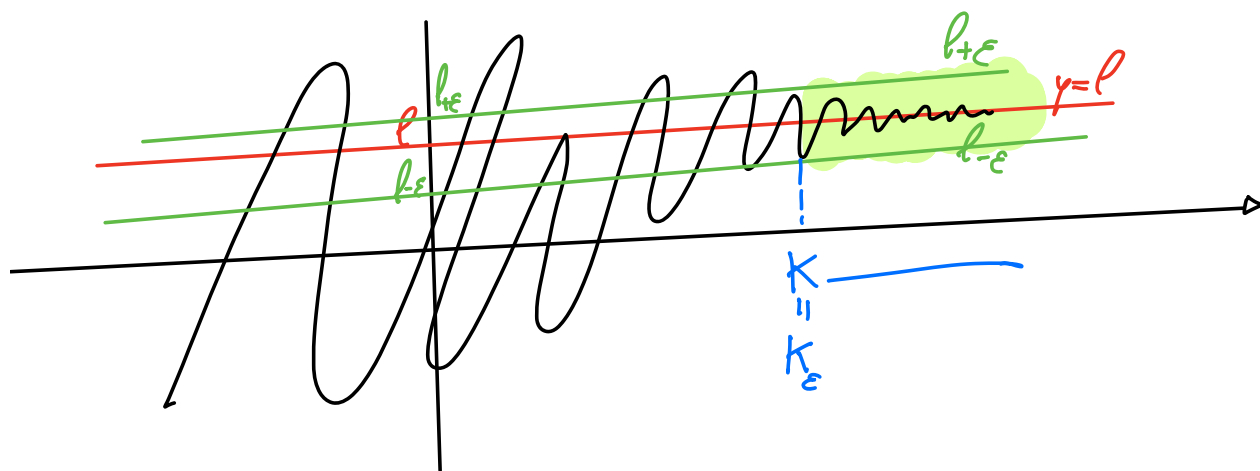
$$\forall I(l \in \mathbb{R}) \quad \exists I(c = +\infty) \quad (\text{danni test } K > 0)$$
$$I_\varepsilon(l) = (l - \varepsilon, l + \varepsilon) \quad \text{"} \quad (K, +\infty)$$

$$\forall \varepsilon > 0 \quad \exists K \in \mathbb{R} \quad (\text{danni test } K > 0)$$

t.c.

$$\forall x \in \text{dom} f \cap \underbrace{I(c = +\infty) \setminus \{c\}}_{(K, +\infty)} \Rightarrow f(x) \in I_\varepsilon(l)$$

$$\forall x \in \text{dom} f, x > K \Rightarrow |f(x) - l| < \varepsilon$$
$$-\varepsilon < f(x) - l < \varepsilon$$
$$l - \varepsilon < f(x) < l + \varepsilon$$



$$\lim_{x \rightarrow -\infty} f(x) = l \in \mathbb{R}$$

$\Rightarrow y=l$ asintoto
orizzontale
sx

$$I(-\infty) = (-\infty, K)$$

$$\forall \varepsilon > 0 \quad \exists K \in \mathbb{R} \text{ (non è restrittivo supporre } K < 0) \text{ t.c.}$$

$$\forall x \in \text{dom} f, x < K \Rightarrow |f(x) - l| < \varepsilon$$

(ES)

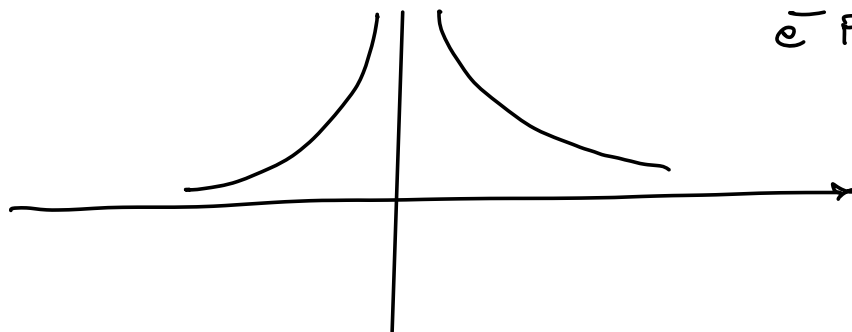
$$f(x) = \frac{1}{|x|}$$

$$\text{dom} f = \mathbb{R} \setminus \{0\}$$

$\bar{c}$ continua (ritornando $\forall x \in \text{dom} f$)

$\bar{c}$ pari

$$f(-x) = f(x) \\ \forall x \in \text{dom} f$$



$$\lim_{x \rightarrow 0} f(x) = \lim_{x \rightarrow 0^+} f(x) \stackrel{\text{PARI}}{=} \lim_{x \rightarrow 0^-} f(x) = +\infty$$

$\Rightarrow x=0$ asintoto verticale completo

$$\lim_{x \rightarrow +\infty} f(x) \stackrel{\text{PARI}}{=} \lim_{x \rightarrow -\infty} f(x) = 0$$

$\Rightarrow y=0$ asintoto orizzontale completo

FATE VOI I 4 CASI : $C = \pm \infty$, $L = \pm \infty$

ES

$$f(x) = \sin \frac{1}{x}$$

È DISPARI $f(-x) = -f(x)$
 $\text{dom} f = \mathbb{R} \setminus \{0\}$

$$\lim_{x \rightarrow x_0} f(x) = f(x_0) = \sin \frac{1}{x_0} \quad \forall x_0 \in \text{dom} f$$

f è continua

HA SENSO CHIEDERSI SE $\exists$

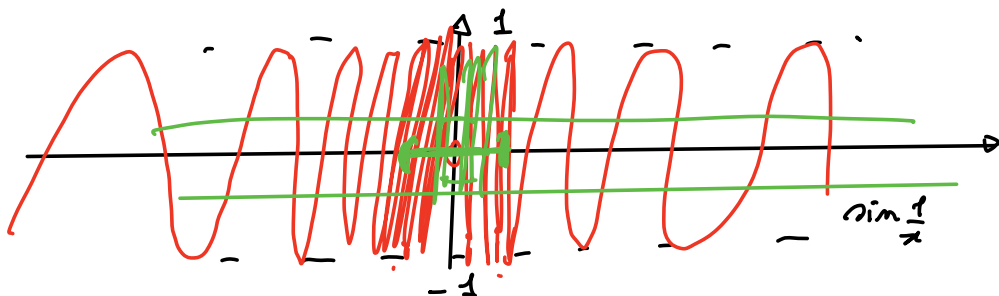
$\downarrow$

$\begin{matrix} +\infty \\ -\infty \\ 0 \end{matrix} \int$ di acc.
per dom f

$$\lim_{x \rightarrow +\infty} \sin \frac{1}{x} = 0$$

$$\lim_{x \rightarrow -\infty} \sin \frac{1}{x} = 0$$

$$\lim_{x \rightarrow 0} \sin \frac{1}{x} \quad ?$$



$$\lim_{x \rightarrow 0} \dim \frac{1}{x} \quad \cancel{\square}$$

